

System Wiki Reference Guide

A shareable, editable explanation of how the prototype works: data, rules, treatments, AI guardrails, personas, Live Consult, and operator practices.

Purpose	Help stakeholders review the system without reading a full PRD or source code.
Source	<code>docs/wiki</code> markdown plus <code>src/data/config/wiki-nav.json</code> .
Coverage	9 wiki sections covering Brand Doctor foundations, No Magic AI, and operating rules.
Generated	June 12, 2026 at 8:18 AM

No Magic principle: observed data, deterministic logic, and AI interpretation should stay visibly separate so questions become productive instead of mysterious.

Document Map

Use this as a discussion guide. The Word file is editable for comments, wording changes, and clarification requests; the PDF is for easy read-ahead sharing.

#	Section	What it answers
1	System Overview	What Brand Doctor is, who it serves, and how the end-to-end journey works.
2	Data Model	How Brand Health Records, metrics, trends, occasions, and GN bridge signals fit together.
3	Diagnosis Engine	How deterministic diagnosis rules fire, how evidence is attached, and how confidence is read.
4	Treatment Library	How treatment paths are configured, linked to diagnoses, ranked, and shown as options to test.
5	No Magic AI	What comes from observed data, deterministic logic, and AI interpretation.
6	AI Personas	How the six assistant personalities change tone and reasoning posture without changing facts.
7	Live Consult	How voice, prompt chips, screen actions, and evidence chips work together.
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		interpretation boundaries for BBE, Pricing Power, category lens, and causality.
9	Operator Guide	How to safely update JSON config, validate data, and keep the prototype trustworthy.

System Overview

Source	docs/wiki/00-overview.md
Audience read	What Brand Doctor is, who it serves, and how the end-to-end journey works.

BBE Brand Doctor is a single-brand diagnostic prototype for brand managers and insights leads. It treats the brand as the patient, BBE and Growth Navigator as the test results, deterministic rules as the doctor, and the treatment library as the pharmacy.

The product journey is intentionally sequenced:

1. Start Here
2. Find My Brand
3. Brand Health Panel
4. Current Diagnosis
5. Why We Believe It
6. Root Cause Explorer
7. Dialog With Data
8. Treatment Pharmacy
9. Prescription & Action Plan
10. Follow-Up Signals

The core principle is belief before prescription. A user should understand the health record, believe the diagnosis, inspect the evidence, and only then consider treatment paths to test.

Who It Serves

- Brand managers need a clear read, practical implications, and treatment paths.
- Insights leads need evidence depth, rule trace, caveats, source periods, and guardrails.
- Stakeholders need the system to explain itself without becoming a black box.

What It Is Not

- It is not a two-brand comparison tool.
- It is not AIM run orchestration.
- It is not Supabase-backed yet.
- It is not SKU-level pricing guidance.
- It is not cannibalization, substitution, portfolio migration, or causality proof.

Current Prototype State

The current app is JSON-first. Brand records, diagnosis definitions, rules, treatment definitions, treatment links, personas, prompts, and Live Consult actions all live in local config or data files. React components render. Services compute. Config defines.

Data Model

Source	docs/wiki/01-data-model.md
Audience read	How Brand Health Records, metrics, trends, occasions, and GN bridge signals fit together.

The central contract is the Brand Health Record. Every major product surface reads from this record or from config-driven services derived from it.

Brand Health Record

The Brand Health Record contains:

- Brand identity: `brandId`, `brandName`, `category`, `country`, `period`, `typology`, and `portfolio role`.
- Category lens: `valid uses`, `invalid uses`, and `known blind spots`.
- Core BBE metrics: `Demand Power`, `Pricing Power`, `Meaningful`, `Different`, and `Salient`.
- Metric metadata: `value`, `benchmark band`, `Ahead status`, `momentum`, `source`, `wave`, and `slide`.
- Trends: `lightweight over-time points` for selected metrics.
- Occasions: `occasion-level memory` or `association signals` where available.
- Growth Navigator bridge: `seeded commercial vitals` for brands with available GN extracts.
- Source files: the documents or extracts backing the record.

Config Around The Record

The record does not contain all product logic. The system combines it with config:

- `diagnosis-definitions.json` defines diagnosis meaning and treatment families.
- `diagnosis_rules.json` defines deterministic rule conditions.
- `treatment-definitions.json` defines treatment paths.
- `diagnosis-treatment-links.json` maps diagnoses to treatments.
- `personas.json` defines AI personalities.
- `live-consult-actions.json` defines allowed screen actions.

Source Period Rule

Every interpretation should preserve the source period. A metric from MAT Q126 should not be described as current market truth outside that period. The app should state when Growth Navigator or momentum data is missing rather than filling gaps.

Data View

The `/brand/[brandId]/data` route exposes the source packet in a read-only way. It is the best place to inspect the Brand Health Record, metrics, trends, occasions, GN bridge, diagnosis rules, treatment config, and AI context packet.

Diagnosis Engine

Source	docs/wiki/02-diagnosis-engine.md
Audience read	How deterministic diagnosis rules fire, how evidence is attached, and how confidence is read.

The diagnosis engine is deterministic. AI can explain the result, but it does not choose the diagnosis.

How A Diagnosis Fires

The engine evaluates rules from `diagnosis_rules.json` against the active Brand Health Record. A rule can include:

- Required conditions.
- Any-of conditions.
- Counter-evidence conditions.
- Thresholds on metric value, benchmark band, Ahead status, or momentum.

The highest-supported fired rule becomes the primary diagnosis. If no deterministic rule fires, the seeded Brand Health Record diagnosis can be used as a fallback trace.

Evidence Ledger

For each diagnosis, the system exposes:

- Supporting evidence.
- Complicating evidence.
- Missing evidence.
- What not to conclude.
- Rule confidence.
- Source and slide context where available.

Rule & Evidence Trace

The Rule & Evidence Trace drawer makes the system inspectable. It shows:

- The fired rule.
- Matched conditions.
- Missing or unmatched conditions.
- Counter-evidence.

- All evaluated rules.
- Treatment links.
- Pricing Power and category-lens guardrails.

Confidence

Confidence is not a universal truth score. It is a practical read of how much evidence is present in the current packet. Momentum gaps, missing GN data, or weak benchmark coverage should reduce confidence or make it directional.

Important Boundary

Do not diagnose causality from these rules. A fired diagnosis says the observed pattern fits a configured diagnostic profile. It does not prove why the brand got there.

Treatment Library

Source	docs/wiki/03-treatment-library.md
Audience read	How treatment paths are configured, linked to diagnoses, ranked, and shown as options to test.

The treatment library is the pharmacy. It provides treatment paths to consider or test after the diagnosis is established.

Treatment Definitions

Treatment definitions live in `treatment-definitions.json`. Each treatment includes:

- Name and family.
- Tier.
- Time to impact.
- Cost.
- Difficulty.
- Likelihood.
- Owners.
- Dependencies.
- Expected metric movement.
- Follow-up signals.

Diagnosis Links

`diagnosis-treatment-links.json` maps diagnoses to treatment paths. The link explains why a treatment fits a diagnosis and when not to use it.

Ranking

Treatment ranking is computed in the service layer. The ranking considers:

- Diagnosis-treatment link priority.
- Treatment config.
- Expected metric movement.
- Likelihood.
- Binding-constraint fit.

- Foundation-first logic.

React components do not manually prescribe or reorder treatments. They render the ranked options and reasons.

Prescription Language

Treatment language should remain careful:

- Say "options to consider."
- Say "paths to test."
- Say "draft plan."
- Do not say the system has made the final prescription decision.

The human brand and insights team owns the prescription decision.

Follow-Up Signals

Follow-up signals connect treatment paths back to proof. They tell the team what to watch next quarter and over six to twelve months.

No Magic AI

Source	docs/wiki/04-no-magic-ai.md
Audience read	What comes from observed data, deterministic logic, and AI interpretation.

No Magic means every AI answer should make it clear what came from data, what came from deterministic logic, and what the AI interpreted or drafted.

Three Layers

- 11. Observed data: Brand Health Record metrics, source files, trends, occasions, and GN bridge signals.
- 12. Deterministic logic: diagnosis rules, evidence ledger, treatment links, ranking logic, and guardrails.
- 13. AI interpretation: plain-English summary, translation, challenge, narrative, and meeting-ready framing.

AI should explain and interrogate. It should not invent facts, diagnoses, treatments, competitors, causality, SKU pricing recommendations, cannibalization proof, or portfolio migration proof.

Chat Scope

Dialog With Data is scoped to:

- Active brand.
- Active visual.
- Evidence ledger.
- Rule trace.
- Treatment library.
- Follow-up signals.
- Selected persona.

If a question needs data outside the packet, the assistant should say what is missing.

Why This Matters

Stakeholders will be skeptical because the technology is new and the output is polished. The system should make skepticism productive by exposing citations, rule traces, and caveats in context.

Current Surfaces

- Report sidecar chat renders markdown and scoped answers.
- Brand conversation page includes expandable No Magic citations.
- Live Consult attaches evidence chips to transcript moments.
- Data view exposes the source packet.

AI Personas

Source	docs/wiki/05-personas.md
Audience read	How the six assistant personalities change tone and reasoning posture without changing facts.

AI personas change tone and reasoning posture. They do not change the facts, deterministic diagnosis, treatment library, or guardrails.

The interactive persona selector lives at </wiki/personas>. Use it to compare personas, choose the right lens, and see which solution outputs each persona influences.

Where Personas Show Up

- Executive Summary: changes strategic framing, headline emphasis, and narrative voice.
- Dialog With Data: changes answer posture, tone, and caveat style.
- Brand Conversation: changes how the dedicated conversation page explains or challenges the same scoped data.
- Live Consult: changes spoken reasoning posture while keeping visual actions allowlisted.

Personas do not change deterministic diagnosis, KPI facts, evidence ledgers, treatment rankings, missing-data rules, or guardrails.

Brand Doctor

Balanced diagnostic assistant for general use. It starts with the diagnosis, separates fact from interpretation and caveat, and keeps answers practical.

Best for:

- General diagnosis questions.
- Evidence interpretation.
- Treatment path explanation.

CMO Advisor

Executive strategic advisor. It translates diagnostic evidence into leadership choices, tradeoffs, and focus.

Best for:

- Board or ELT framing.
- Strategic choices.
- Growth priorities.

Insights Skeptic

Pressure-testing persona. It challenges unsupported conclusions while staying useful.

Best for:

- Evidence review.
- Rule logic.
- Missing data.
- What not to conclude.

Creative Strategy Partner

Brand and creative translation persona. It connects diagnosis to proposition, distinctiveness, salience, memory structures, and creative briefing.

Best for:

- Positioning implications.
- Distinctive assets.
- Creative briefing.

RGM / Pricing Partner

Pricing Power guardrail persona. It explains brand-level price justification without turning BBE into SKU pricing guidance.

Best for:

- Pricing Power questions.
- Value justification.
- RGM handoff.

Executive Briefer

Compressed meeting-readout persona. It gives the smallest useful strategic read while preserving evidence and caveat.

Best for:

- Short summaries.
- Stakeholder updates.

- Meeting prep.

Byron Sharp Growth Lens

How Brands Grow-inspired marketing science lens. It brings the spirit of empirical brand growth thinking into Brand Doctor without pretending to be Byron Sharp or claiming official Ehrenberg-Bass endorsement.

It emphasizes:

- Mental availability.
- Physical availability.
- Distinctive brand assets.
- Broad reach across category buyers.
- Penetration and memory-structure building.
- Skepticism toward over-narrow loyalty, niche targeting, or affinity-only reads.

Best for:

- Salience and mental availability questions.
- Distinctive asset implications.
- Broad-reach growth strategy.
- Pressure-testing whether a treatment path is too narrow.

Guardrail:

- This is an interpretive marketing-science lens. It does not change the BBE data, deterministic diagnosis, treatment rankings, or No Magic caveats.

Live Consult

Source	docs/wiki/06-live-consult.md
Audience read	How voice, prompt chips, screen actions, and evidence chips work together.

Live Consult is the voice-plus-proof experience. The user can sit back, ask questions, and watch the system move through the report.

What Makes It Work

The experience is not just speech. It combines:

- A brand-scoped voice session.
- Prompt chips for demo-safe testing.
- Browser voice fallback.
- A premium doctor presence.
- Allowlisted screen actions.
- Transcript rendering.
- Evidence chips.
- No Magic context.

Allowed Screen Actions

Live Consult can only control the UI through configured actions in `live-consult-actions.json`.

Examples:

- Open Executive Summary.
- Open Current Diagnosis.
- Open Rule & Evidence Trace.
- Highlight KPI modules.
- Open Prescription & Action Plan.
- Select the top treatment path.
- Create a meeting takeaway stub.

The assistant cannot mutate source data, invent treatments, or bypass deterministic diagnosis logic.

Demo Control

The Demo Control band gives presenters:

- Showcase brands.
- Reset demo.
- Best prompts to try.

Reset returns the prototype to a known-good state: Siete, Insights Lead mode, top of page, Live Consult closed.

Evidence Chips

Live Consult transcript moments attach chips such as:

- Brand.
- Diagnosis.
- Rule.
- Source.
- Metric.
- Treatment.

This keeps the voice interaction grounded and makes the "prove it" moment visible.

Voice Key

The OpenAI API key stays server-side. The browser receives a short-lived Realtime client secret from `/api/live-consult/session`.

Guardrails

Source	docs/wiki/07-guardrails.md
Audience read	The non-negotiable interpretation boundaries for BBE, Pricing Power, category lens, and causality.

Guardrails are part of the product, not legal fine print. They should be visible wherever the user might over-read the data.

Pricing Power

Pricing Power is broad brand-equity price justification. It is not SKU-level pricing guidance.

Do not use Pricing Power to recommend:

- SKU price changes.
- Pack architecture.
- Promo depth.
- Retailer-specific price moves.

Those decisions require RGM, pricing science, and commercial evidence outside the BBE packet.

Category Lens

The active category lens defines the benchmark context. A brand may be read inside Snacks or Better-for-You Snacks, but that does not prove substitution or portfolio migration.

No Causality Claims

The system can identify patterns and diagnostic fit. It cannot prove causality without causal evidence.

No Cannibalization Or Substitution Proof

The current packet does not prove cannibalization, occasion-level substitution, or consumer migration across brands.

No One-Wave Overreach

Do not call a brand permanently miscast based on one wave or one metric. Momentum gaps should be called out as missing or directional.

AI Guardrail

AI can summarize, translate, challenge, and draft. It cannot invent unsupported facts, metrics, treatments, or conclusions.

Operator Guide

Source	docs/wiki/08-operator-guide.md
Audience read	How to safely update JSON config, validate data, and keep the prototype trustworthy.

This prototype is intentionally JSON-first. Most changes should happen in config files, then pass validation before anyone demos the app.

Common Files

- Brand records: `src/data/demo/brand-health-records.json`
- Diagnosis definitions: `src/data/config/diagnosis-definitions.json`
- Diagnosis rules: `src/data/config/diagnosis_rules.json`
- Treatments: `src/data/config/treatment-definitions.json`
- Diagnosis-treatment links: `src/data/config/diagnosis-treatment-links.json`
- Personas: `src/data/config/personas.json`
- Live Consult actions: `src/data/config/live-consult-actions.json`
- Live Consult scenarios: `src/data/config/live-consult-scenarios.json`
- Wiki nav: `src/data/config/wiki-nav.json`
- Wiki content: `docs/wiki/*.md`

Safe Update Flow

14. Edit JSON or markdown source.
15. Run `pnpm validate:data`.
16. Run `pnpm typecheck`.
17. Run `pnpm build`.
18. Browser-test the affected route.
19. Update `STATUS.md` and `BACKLOG.md`.
20. Commit and push to the private GitHub repo.

Config Rule

Diagnosis and treatment logic should come from config and service-layer computation. React components should render computed/configured content, not hide business logic.

Wiki Exports

The shareable Word and PDF guide is generated from the same markdown files that power `/wiki`.

- Source docs live in `docs/wiki`.
- Navigation and summaries live in `src/data/config/wiki-nav.json`.
- The Word export lives at `public/exports/bbe-brand-doctor-system-wiki.docx`.
- The PDF export lives at `public/exports/bbe-brand-doctor-system-wiki.pdf`.
- Regenerate the Word file with `scripts/export-wiki-docs.py`, then render it to PDF with the document renderer before sharing.

The Word file is meant for team edits, comments, and clarification requests. The PDF is meant for read-ahead sharing when people should not edit the artifact directly.

What To Avoid

- Do not add Supabase until the JSON-first contract is stable.
- Do not bring back AIM run framing.
- Do not turn the app into a two-brand comparison tool.
- Do not put API keys or local secrets in Git.
- Do not overwrite source data without validation and preview.

Future Editing Workflow

When config editing is added, it should use:

- Draft changes.
- Impact preview.
- Accept or reject.
- Batch acceptance.
- Version history.
- Rollback/export.